



UNIVERSITY OF SRI JAYEWARDENEPURA

Faculty of Technology

Department of Information and Communication Technology

ITS 4202 - Emerging Technologies

Assignment 1

Designing and Modeling a Smart Agriculture Solution

Name: K.P.V. PERERA

Index Number: ICT/21/897

Automated Fertigation System for Corn Farming in Monaragala

Table of Contents

1.Introduction.....	3
2.Farm Details and Situation	3
2.1 Farm Details:-	3
2.2 Current Situation:-	3
3. Key challenges	3
4. Technology Research	4
4.1 Soil Moisture Sensors	4
4.2 NPK Sensors	4
4.3 Fertigation Technology	4
4.4 Drip Irrigation	4
4.5 IoT and Cloud Computing	4
5. System design	5
5.1 System Components.....	5
5.1.1 Sensors used in the field.....	5
5.1.2 Irrigation Equipment.....	5
5.1.3 Communication System.....	5
5.1.4 Software	5
5.2 Why do These Components need to be used?.....	6
5.3 Conceptual Model	6
6. Implementation plan and cost analysis	7
7. Cost Analysis	8
8. How the system works	9
9. Potential challenges and solutions	10
9.1 Small farmers don't have LKR 700,000-880,000 amount of money	10
9.2 Farmers don't know how to fix technical issues	10
9.3 Farm data could be stolen by hackers	10
9.4 Sensors might give wrong readings	10
10. Conclusion	11
11.References.....	11

1.Introduction

Corn is the second most important cereal crop in Sri Lanka. Cultivated seasonally in the dry zone during the Maha and Yala seasons, mainly in the Anuradhapura, Polonnaruwa and Monaragala districts.

Monaragala is the second largest district in Sri Lanka. The total land area is 5,636 square kilometers. It is a part of a Dry Zone. Specially noted as a high drought affected and low rainfall area. Farmers face challenges like water scarcity and fertilizer shortages .This contributes to reduce the corn yield. As a solution for this case,

This report shows a smart farming solution for corn farmers in Monaragala district. This smart system combines water irrigation and fertilizer application together. The system uses sensors, cloud computing, and automatic controls. It can save 40% water and 30% fertilizer costs. It also increases corn production by 20-25%.

2.Farm Details and Situation

2.1 Farm Details:-

- **Farm Type:** Corn Farming
- **Location:** Monaragala District (Dry Zone)
- **Farm Size:** 5-10 acres
- **Climate:** Very dry, low rainfall

2.2 Current Situation:-

Monaragala is one of the driest places in Sri Lanka. Corn farmers don't have enough water. The dry season is very long (5-7 months). Traditional farming methods waste a lot of water and fertilizer. Farmers don't know when to water plants or add fertilizer. This causes low crop production and wasted money.

3. Key challenges

- **Water Scarcity**

Monaragala gets very little rain each year (only 800-1000mm). The dry season is very long. Underground water is running out. Old irrigation methods waste 50-60% of water. When corn plants

need water most (during flowering), water shortage reduces production by 30-40%. Farmers don't have scientific information about when to water.

- **Fertilizer Wastage**

Farmers throw fertilizer by hand. This method is very inefficient. Only 30-40% of fertilizer reaches plant roots. The rest is wasted. Farmers use too much fertilizer because they don't know soil nutrient levels. This wastes money and damages soil. Extra fertilizer pollutes water sources. Without proper timing, plants cannot use the fertilizer properly.

4. Technology Research

4.1 Soil Moisture Sensors

These sensors measure water in soil. Research shows sensor based irrigation saves 30-50% water. For corn, soil should have 60-70% moisture during important growth stages.

4.2 NPK Sensors

NPK sensors measure three important nutrients: Nitrogen (N), Phosphorus (P), and Potassium (K). These sensors help farmers know exactly what nutrients soil needs. Studies show NPK sensors reduce fertilizer use by 25-35%.

4.3 Fertigation Technology

Fertigation means adding fertilizer through irrigation water. A special machine mixes fertilizer into water. This delivers nutrients directly to plant roots with 85-95% efficiency. Traditional methods only have 30-40% efficiency.

4.4 Drip Irrigation

Drip irrigation gives water directly to plant roots through pipes. This method is 90-95% efficient. Old flood irrigation is only 40-50% efficient. Drip systems reduce water loss from evaporation.

4.5 IoT and Cloud Computing

IoT connects sensors and mobile phones through internet. Cloud computing stores and processes data on remote servers. Farmers can check their fields from anywhere using smartphones. AI can analyze data and predict best times for irrigation and fertilization.

5. System design

5.1 System Components

5.1.1 Sensors used in the field

- Soil Moisture Sensors to Measure water levels in soil (6 units)
- NPK Sensors to Measure fertilizer nutrients in soil (3 units)
- Soil pH Sensor to Check soil acidity (1 unit)
- Weather Station to measure temperature, humidity, and rainfall (1 unit)

5.1.2 Irrigation Equipment

- **Drip Irrigation System:** Pipes and drippers for the entire farm
- **Fertilizer Injector:** A Machine to mix fertilizer into water
- **Automated Valves:** Control water and fertilizer flow
- **Water Pump:** Pump water from a well or a tank
- **Filters:** Clean water to prevent pipe blocking

5.1.3 Communication System

- **LoRaWAN Gateway:** Wireless system to connect sensors (works up to 5km)
- **Solar Panels** to provide power using sunlight
- **Controller:** This is a small computer to manage everything.

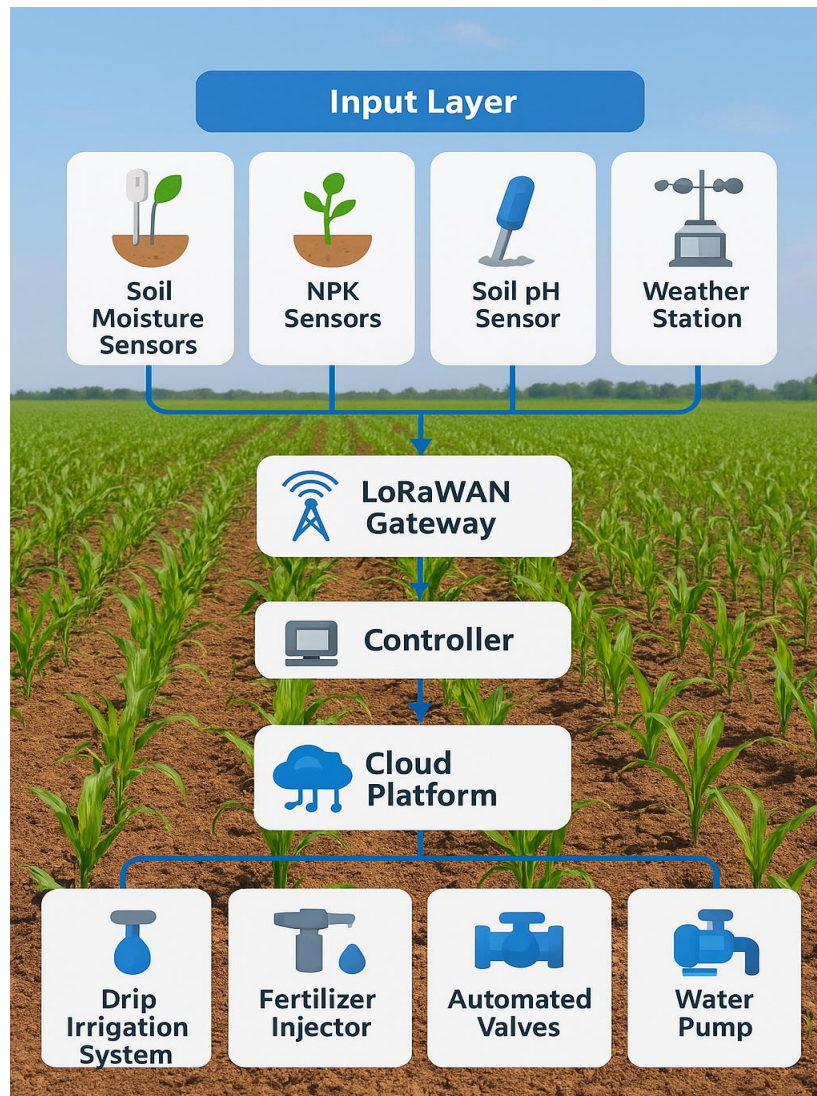
5.1.4 Software

- **Cloud Platform:** Store all data online
- **Mobile App:** Sinhala language app for farmers
- **AI System:** Smart System to make irrigation decisions

5.2 Why do These Components need to be used?

- **Soil Moisture Sensors:** Accurate, long-lasting (3-5 years), affordable (LKR 6,000 each)
- **Fertilizer Injector:** Works without electricity. More reliable for rural areas.
- **LoRaWAN:** Works long distance, uses very little power, works without phone signal.
- **Drip Irrigation:** Most efficient. Saves 50% water. Pays for itself in 2-3 years.
- **Cloud Platform:** Safe data storage. Farmers can check from anywhere.

5.3 Conceptual Model



6. Implementation plan and cost analysis

Phase 1: Installation (Month 1-2)

- Test the soil to know nutrient levels
- Design drip irrigation layout
- Install all pipes and drippers
- Install water pump and fertilizer injector
- Install automated valves

Phase 2: Sensors (Week 3-4)

- Install soil moisture sensors
- Install NPK sensors
- Install weather station
- Install solar panels
- Install wireless gateway

Phase 3: Software Setup (Week 5-6)

- Setup cloud platform
- Create database for storing data
- Program controller
- Develop mobile app in Sinhala
- Setup alert system

Phase 4: Testing (Week 7-8)

- Test all sensors
- Check if readings are correct
- Test irrigation system
- Test fertilizer mixing
- Test mobile app

Phase 5: Training (Week 9-10)

- Train farmer to use mobile app
- Explain how system works
- Show how to do basic maintenance
- Provide Sinhala user manual
- Give technical support contact

Phase 6: Start Using (Month 3)

- Plant corn with system running
- Watch system daily for first month
- Collect data for AI learning
- Make small adjustments
- Record water and fertilizer savings

7. Cost Analysis

Item	Quantity	Price (Estimated)
Drip irrigation system	1 set	250,000 - 400,000
Soil moisture sensors	6 units	36,000
NPK sensors	3 units	90,000
Weather station	1 unit	45,000
Automated valves	3 units	30,000
Fertilizer injector	1 unit	25,000
Communication system	1 set	40,000
Solar panels	1 set	60,000
Software & app	1 year	105,000
Installation	-	50,000

Total Cost	731,000 - 881,000
-------------------	--------------------------

Annual Running Cost: LKR 50,000 (cloud services, maintenance)

Annual Savings:

- Water savings: LKR 120,000
- Fertilizer savings: LKR 90,000
- Higher yield income: LKR 200,000
- Labor savings: LKR 60,000

Total benefit: LKR 470,000 per year

System pays for itself in 1.5 to 2 years

8. How the system works

Step 1

Sensors measure soil moisture, nutrients, and weather every 30 minutes



Step 2

Data is sent wirelessly to cloud computer



Step 3

AI analyzes data and calculates water and fertilizer needs



Step 4

System opens valves and activates fertilizer injector automatically



Step 5

Farmer gets updates on mobile phone

9. Potential challenges and solutions

9.1 Small farmers don't have LKR 700,000-880,000 amount of money

Solutions:

- Get agricultural loans from banks
- Form farmer groups to share costs (10 farmers together)
- Start simple, add features later
- Apply for government support programs

9.2 Farmers don't know how to fix technical issues

Solutions:

- Partner with agricultural extension office
- Train one family member as system operator
- Provide remote help through mobile app
- Create maintenance contracts with technicians
- Design manual backup operation

9.3 Farm data could be stolen by hackers

Solutions:

- Use encrypted communication
- Strong passwords with two-step login
- Secure cloud servers with backups
- Only farmer can access data

9.4 Sensors might give wrong readings

Solutions:

- Use multiple sensors and average readings
- Calibrate sensors every 6 months
- Use high-quality waterproof sensors
- Check manually sometimes to verify

10. Conclusion

This automated fertigation system solves major problems of corn farmers in Monaragala. The system combines drip irrigation with precise fertilizer application. It is controlled by sensors and smart algorithms. It addresses water scarcity and fertilizer waste effectively.

The system's main strengths are precision, automation, and smart decisions. Farmers don't need to guess. The system monitors soil continuously. It applies water and nutrients exactly when needed and in right amounts. This saves money and increases production.

The system can be installed gradually. Start with basic drip irrigation. Add automation later. This makes it suitable for different budgets. The system works for other crops too, not just corn. Beyond money benefits, this solution helps environment by saving water and reducing pollution. In dry areas like Monaragala, such technologies are essential for long term farming success and food security.

11. References

<https://www.researchgate.net>

<https://www.automation.com>

<https://www.automationworld.com>

<https://www.cotroleng.com>

<https://www.rockwellautomation.com>

<https://www.siemens.com>

<https://www.ibm.com>

<https://www.britannica.com>

<https://www.conger.com>

<https://www.agromini.com>

<https://www.sciencedirect.com>

<https://www.agro.elzian.com>